EDAN LIAHOVETSKY

Robotics Software | Autonomous Systems | Open-Source Engineering

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PROFILE

President and programming lead of Rebel Robotics, a 100+ member FIRST Robotics Competition (FRC) team. Created BLine, an open-source autonomy suite adopted by 50+ FRC teams, including 24 at the 2026 World Championship.

EDUCATION

Great Neck South High School | Great Neck, NY Class of 2027

GPA: 98.6 weighted / 93.9 unweighted **SAT:** 1540 superscore (790 Math, 750 Reading & Writing)

SELECTED ENGINEERING PROJECTS

BLine | Creator, Maintainer & Community Lead Jun 2025–Present

- Identified limited robot testing time as an autonomy-development bottleneck; designed BLine to help teams author, tune, and validate reliable routines despite imperfect drivetrains.
- Built endpoint path tracking, cross-track correction, handoff radii, motion constraints, Java/WPILib integration, and web/desktop path-authoring and simulation tools.
- Adopted by 50+ teams, including 24 at the 2026 World Championship. In 2026, adopting teams won 12 competitions (2 regionals, 5 district events, 5 state/district championships) and a World Championship division; one ranked #1 in its Championship division and won its Autonomous Award.
- **Working with FIRST developers** on BLine integration into the official FRC software ecosystem, targeting the 2027 season.
- In randomized WPILib physics-simulation trials against PathPlanner using an Artificial Bee Colony optimizer, measured 97% lower path computation time and 66% lower waypoint cross-track error.
- Maintain project documentation, releases, and direct technical support for adopting teams.

wpilib-agent-tools | Creator Feb 2026–Present

- Architected an open-source toolkit enabling AI coding agents (Codex, Claude Code, Cursor) to run robot simulations, collect telemetry, analyze logs, revise code, and re-test in sandboxed engineering loops.
- Shared the toolkit with the FRC community and provided user support; gained practical experience building engineering workflows around AI coding agents.

ROBOTICS LEADERSHIP & SYSTEMS ENGINEERING

Rebel Robotics, FRC Team 2638 | President & Programming Lead Member since 2022

- Lead all-hands meetings and competition strategy for a 100+ member team; recruit and mentor new members through hands-on technical workshops; present the team's work at public events.
- Rebuilt the Java/WPILib software stack around structured logging, log replay, and physics simulation; led swerve-drive, vision-based pose estimation, and automated alignment development.
- Coach the drive team on autonomous selection, pre-match checks, and field debugging; operate and debug the robot at events, including four FIRST World Championship qualifications. Team earned two 2026 Excellence in Engineering Awards and the 2024 World Championship Engineering Inspiration Award.

RESEARCH & ENGINEERING EXPERIENCE

NASA Glenn High School Engineering Institute | July 2025. Completed hands-on engineering projects, received Outstanding Achievement recognition, and arranged additional rover-mobility laboratory shadowing.

Applied research. Built geospatial datasets and a Random Forest model to study natural hydrogen occurrence and model limitations; generated synthetic imagery in Unreal Engine 5 for drone-based litter-detection research.

TECHNICAL SKILLS

Programming: Java (primary); Python and C++ (working knowledge); TypeScript, React, and Tauri (project experience).
Robotics & systems: WPILib, PID/motion profiling, computer vision, AprilTags, physics simulation, CAN/CAN FD.
Hardware: Robot wiring and electrical integration, 3D printing; working knowledge of CAD and CNC.

HONORS & ADDITIONAL ACTIVITIES

Honors: National Merit Commended Student (2026); AP Scholar with Distinction (2026).
Additional: Paid FIRST LEGO League coach; founder and pianist, Flat 7 jazz-fusion quartet.